

Down-and-out put — traitements du coin : figures

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Recueil des figures et tables produites par les scripts d'agrégation, sans aucun recalcul : chaque légende indique ce qui est tracé ; les chemins des artefacts sources sont donnés en tête de section. Notation : $\Delta = K - B$, V_{DOD} le prix du digital down-and-out, $\xi = \ln(s/B)/(\sigma\sqrt{2(T-t)})$, χ le cutoff de l'enrichissement, $N_{0.1} = \{|s - B| + (T - t) \leq 0.1\}$ la fenêtre de coin des métriques.

1 Comparaison entre configurations (médianes sur les graines)

Chaque point est une graine maîtresse, le losange plein la médiane. Les configurations de lissage excluent la fenêtre de coin $N_{0.1}$ de la collocation ; les traitements analytiques (soustraction, enrichissement) incluent le coin, ce que l'étiquette de l'axe rappelle. Source : `data/aggregate_terminal_function_comparison/20260921_corner_treatments_iters50000`.

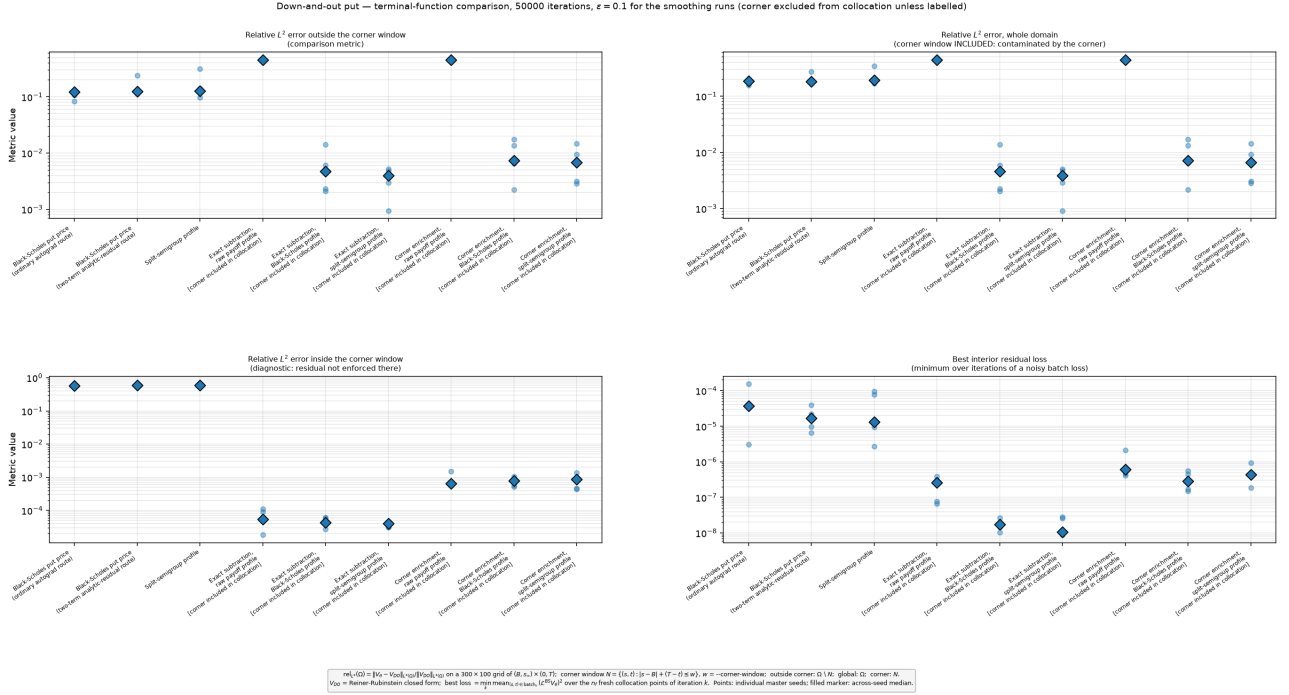


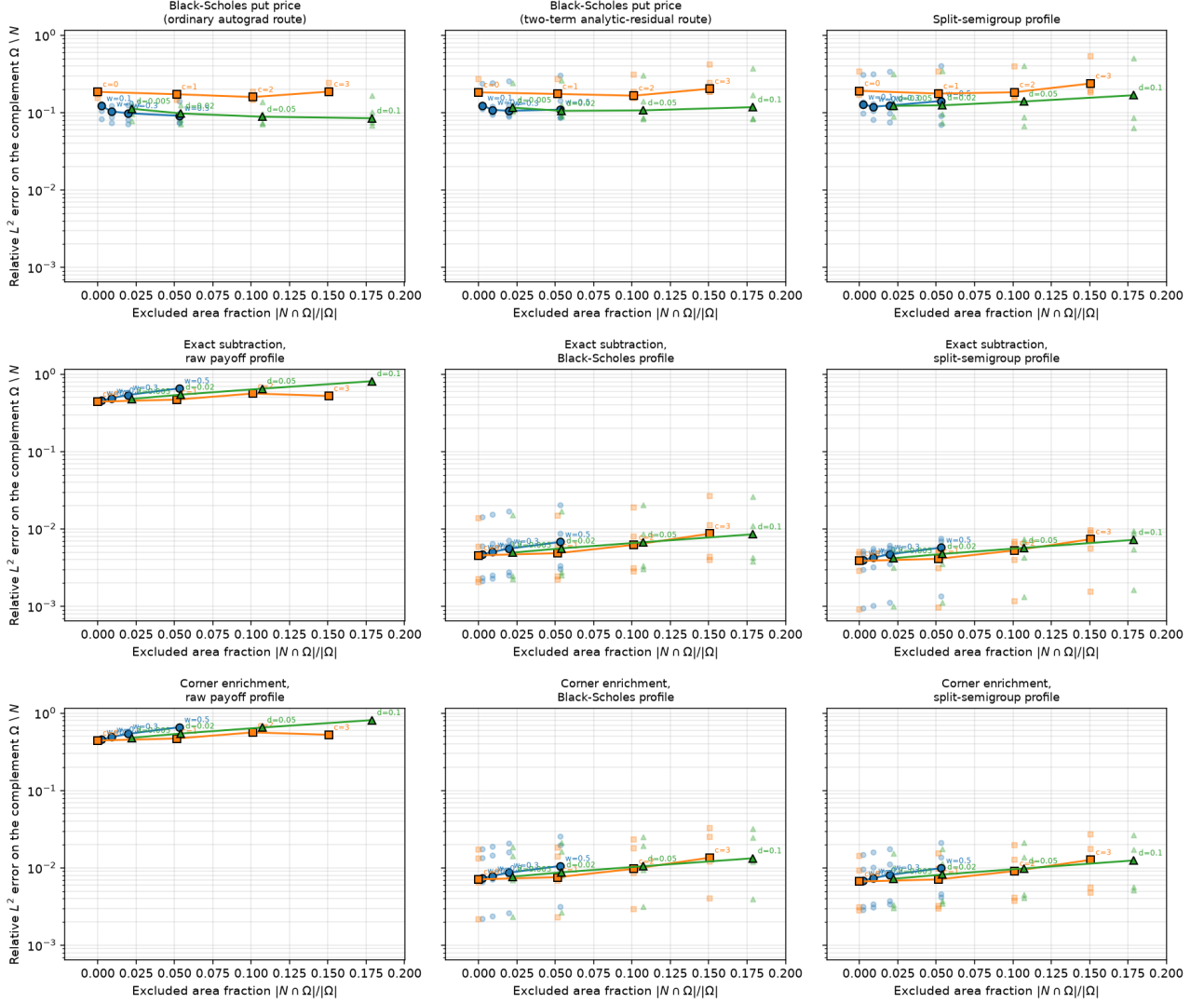
FIGURE 1 – Erreur relative L^2 hors fenêtre de coin (métrique de comparaison), sur tout le domaine, dans la fenêtre, et meilleure perte intérieure ; 50000 itérations, $\varepsilon = 0.1$ pour les runs de lissage.

Configuration	n	'rel_l2_outside_corner'	'rel_l2_global'	'rel_l2_corner'	'max_abs_error_outside_corner'	'best_loss'	'best_iter'
Black-Scholes put price (ordinary autograd route)	5	1.218e-01 [8.310e-02, 1.305e-01]	1.855e-01 [1.548e-01, 1.900e-01]	5.837e-01 [5.553e-01, 6.080e-01]	9.555e-02 [6.732e-02, 1.098e-01]	3.675e-05 [3.035e-06, 1.548e-04]	48150 [46551, 49611]
Black-Scholes put price (two-term analytic-residual route)	5	1.238e-01 [1.125e-01, 2.373e-01]	1.827e-01 [1.789e-01, 2.722e-01]	5.958e-01 [5.678e-01, 6.094e-01]	1.017e-01 [8.351e-02, 1.120e-01]	1.671e-05 [6.437e-06, 3.915e-05]	48335 [45215, 49923]
Split-semigroup profile	5	1.270e-01 [9.763e-02, 3.113e-01]	1.911e-01 [1.681e-01, 3.404e-01]	5.864e-01 [5.628e-01, 6.576e-01]	9.760e-02 [8.040e-02, 1.467e-01]	1.299e-05 [2.730e-06, 9.448e-05]	49474 [48150, 49879]
Exact subtraction, raw payoff profile	5	4.549e-01 [4.537e-01, 4.559e-01]	4.418e-01 [4.407e-01, 4.428e-01]	5.381e-05 [1.812e-05, 1.093e-04]	1.176e-01 [1.174e-01, 1.177e-01]	2.581e-07 [6.530e-08, 3.878e-07]	47907 [46834, 48332]
Exact subtraction, Black-Scholes profile	4	4.687e-03 [2.116e-03, 1.422e-02]	4.552e-03 [2.055e-03, 1.381e-02]	4.335e-05 [2.693e-05, 6.171e-05]	2.224e-03 [7.103e-04, 5.538e-03]	1.711e-08 [1.044e-08, 2.687e-08]	49751 [33591, 49974]
Exact subtraction, split-semigroup profile	5	3.968e-03 [9.426e-04, 5.193e-03]	3.854e-03 [9.156e-04, 5.044e-03]	4.051e-05 [3.062e-05, 4.418e-05]	1.226e-03 [2.327e-04, 1.928e-03]	1.057e-08 [8.780e-09, 2.868e-08]	49470 [48857, 49859]
Corner enrichment, raw payoff profile	4	4.557e-01 [4.543e-01, 4.580e-01]	4.426e-01 [4.412e-01, 4.449e-01]	6.418e-04 [5.626e-04, 1.501e-03]	1.177e-01 [1.174e-01, 1.184e-01]	6.160e-07 [4.150e-07, 2.090e-06]	45156 [39595, 49049]
Corner enrichment, Black-Scholes profile	5	7.316e-03 [2.216e-03, 1.764e-02]	7.108e-03 [2.160e-03, 1.713e-02]	7.855e-04 [5.115e-04, 1.058e-03]	2.979e-03 [3.960e-04, 7.044e-03]	2.849e-07 [1.490e-07, 5.603e-07]	49124 [20057, 49920]
Corner enrichment, split-semigroup profile	5	6.854e-03 [2.889e-03, 1.473e-02]	6.658e-03 [2.813e-03, 1.431e-02]	8.729e-04 [4.331e-04, 1.350e-03]	2.900e-03 [4.084e-04, 5.850e-03]	4.405e-07 [1.869e-07, 9.281e-07]	45456 [26131, 49224]

TABLE 1 – Métriques par configuration, médiane [min, max] sur les graines (table.md).

2 Diagnostics à partir des modèles sauvegardés

Down-and-out put — error vs. excluded area for three window shapes, 50000 iterations, $\varepsilon = 0.1$, corner window excluded from collocation



Excluded window family (shape)
 — Lozenge $N_w = \{|s - B| + \tau \leq w\}$ — Parabola $N_c = \{|s - B| \leq cB\sqrt{\tau}\}$ — Hyperbola $N_d = \{(\tau(s - B) \leq d)\}$

$\text{rel}_{L^2}(\Omega \setminus N) = \|V_0 - V_{\text{col}}\|_{L^2(\Omega \setminus N)} / \|V_{\text{col}}\|_{L^2(\Omega \setminus N)}$ on the pilot's 300×100 grid of $\Omega = (B, s_*) \times (0, T)$, $\tau = T - t$: abscissa: $|N \cap \Omega|/|\Omega|$ (grid fraction).
 $N_w = \{|s - B| + \tau \leq w\}$, $w \in \{0.1, 0.2, 0.3, 0.5\}$; $N_c = \{|s - B| \leq cB\sqrt{\tau}\}$, $c \in \{0.1, 2, 3\}$; $N_d = \{(\tau(s - B) \leq d)\}$, $d \in \{0.005, 0.02, 0.05, 0.1\}$. Faint points: individual seeds; solid line: across-seed median; labels: parameter value.

FIGURE 2 – Erreur relative L^2 sur le complémentaire d'une fenêtre exclue autour du coin, pour trois formes de fenêtre (losange ℓ^1 , parabole, hyperbole), en fonction de la fraction d'aire exclue.

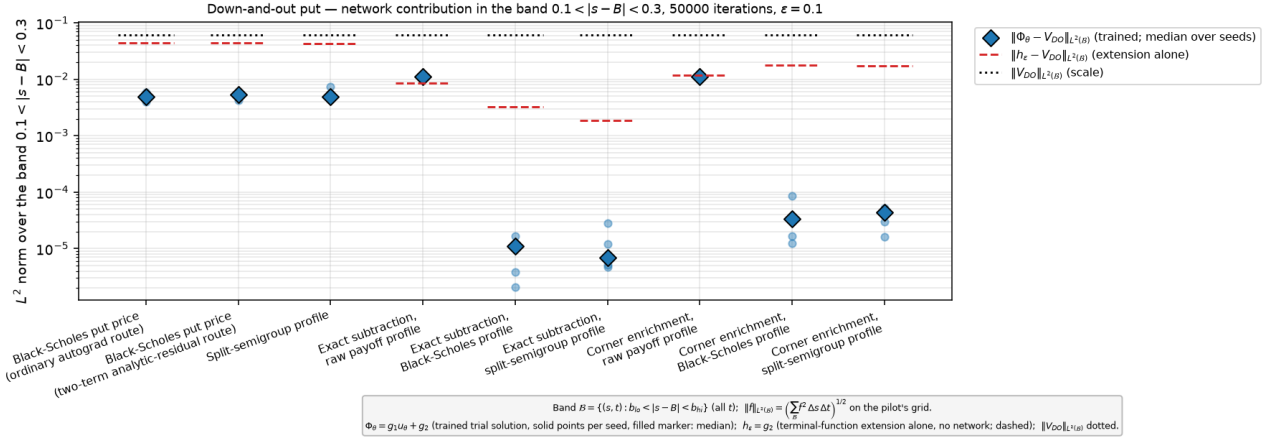


FIGURE 3 – Bande $0.1 < |s - B| < 0.3$: normes L^2 de $\Phi_\theta - V_{DO}$ (solution entraînée) et de $g_2 - V_{DO}$ (extension seule, sans réseau).

Configuration	[0.6, 0.7]		[0.7, 1]		[1, 2]		[2, 3]	
Black-Scholes put price (ordinary autograd route)	2.038e-01	[1.457e-01, 2.402e-01]	7.673e-02	[6.334e-02, 9.088e-02]	5.938e-02	[4.649e-02, 7.725e-02]	1.335e+01	[4.184e+00, 5.720e+01]
Black-Scholes put price (two-term analytic-residual route)	2.161e-01	[1.766e-01, 2.408e-01]	8.333e-02	[6.647e-02, 9.022e-02]	5.104e-02	[3.956e-02, 9.379e-02]	3.335e+01	[1.031e+01, 1.359e+02]
Split-semigroup profile	2.047e-01	[1.702e-01, 2.982e-01]	7.517e-02	[6.902e-02, 1.138e-01]	4.930e-02	[4.216e-02, 8.478e-02]	5.675e+01	[5.935e+00, 1.823e+02]
Exact subtraction, raw payoff profile	3.962e-02	[3.929e-02, 4.073e-02]	3.397e-01	[3.389e-01, 3.409e-01]	1.294e+00	[1.291e+00, 1.294e+00]	6.964e+00	[2.612e+00, 9.760e+00]
Exact subtraction, Black-Scholes profile	1.406e-04	[4.359e-05, 1.708e-04]	1.831e-04	[4.396e-05, 2.910e-04]	1.115e-03	[6.156e-04, 1.806e-03]	3.162e+00	[1.429e+00, 9.626e+00]
Exact subtraction, split-semigroup profile	1.002e-04	[8.625e-05, 3.701e-04]	1.190e-04	[8.630e-05, 4.891e-04]	5.204e-04	[3.707e-04, 1.733e-03]	2.686e+00	[4.651e-01, 3.513e+00]
Corner enrichment, raw payoff profile	3.947e-02	[3.912e-02, 4.072e-02]	3.404e-01	[3.391e-01, 3.427e-01]	1.296e+00	[1.292e+00, 1.300e+00]	2.429e+00	[1.426e+00, 8.814e+00]
Corner enrichment, Black-Scholes profile	4.394e-04	[2.012e-04, 1.088e-03]	5.836e-04	[2.333e-04, 1.345e-03]	1.652e-03	[1.563e-03, 2.709e-03]	4.945e+00	[1.372e+00, 1.193e+01]
Corner enrichment, split-semigroup profile	5.251e-04	[2.606e-04, 1.195e-03]	8.443e-04	[2.944e-04, 1.164e-03]	2.998e-03	[1.120e-03, 8.337e-03]	4.636e+00	[1.044e+00, 9.939e+00]

TABLE 2 – Erreur relative L^2 par bande de s (tout t , fenêtre de coin retirée), médiane [min, max] sur les graines.

Configuration	[0.6, 0.7]		[0.7, 1]		[1, 2]		[2, 3]	
Black-Scholes put price (ordinary autograd route)	6.138e-03	[4.387e-03, 7.232e-03]	5.111e-03	[4.219e-03, 6.054e-03]	1.193e-03	[9.339e-04, 1.552e-03]	1.494e-03	[4.682e-04, 6.400e-03]
Black-Scholes put price (two-term analytic-residual route)	6.509e-03	[5.320e-03, 7.250e-03]	5.551e-03	[4.428e-03, 6.010e-03]	1.025e-03	[7.948e-04, 1.884e-03]	3.731e-03	[1.154e-03, 1.521e-02]
Split-semigroup profile	6.164e-03	[5.125e-03, 8.981e-03]	5.008e-03	[4.598e-03, 7.581e-03]	9.903e-04	[8.471e-04, 1.703e-03]	6.350e-03	[6.641e-04, 2.039e-02]
Exact subtraction, raw payoff profile	1.193e-03	[1.183e-03, 1.227e-03]	2.263e-02	[2.258e-02, 2.271e-02]	2.599e-02	[2.593e-02, 2.601e-02]	7.793e-04	[2.923e-04, 1.092e-03]
Exact subtraction, Black-Scholes profile	4.234e-06	[1.313e-06, 5.143e-06]	1.219e-05	[2.929e-06, 1.938e-05]	2.240e-05	[1.237e-05, 3.629e-05]	3.539e-04	[1.599e-04, 1.077e-03]
Exact subtraction, split-semigroup profile	3.019e-06	[2.597e-06, 1.115e-05]	7.929e-06	[5.749e-06, 3.258e-05]	1.045e-05	[7.447e-06, 3.482e-05]	3.006e-04	[5.205e-05, 3.931e-04]
Corner enrichment, raw payoff profile	1.189e-03	[1.178e-03, 1.226e-03]	2.268e-02	[2.259e-02, 2.283e-02]	2.604e-02	[2.596e-02, 2.612e-02]	2.718e-04	[1.596e-04, 9.863e-04]
Corner enrichment, Black-Scholes profile	1.323e-05	[6.058e-06, 3.275e-05]	3.888e-05	[1.554e-05, 8.960e-05]	3.319e-05	[3.140e-05, 5.443e-05]	5.533e-04	[1.535e-04, 1.335e-03]
Corner enrichment, split-semigroup profile	1.581e-05	[7.846e-06, 3.597e-05]	5.624e-05	[1.961e-05, 7.755e-05]	6.022e-05	[2.249e-05, 1.675e-04]	5.187e-04	[1.168e-04, 1.112e-03]

TABLE 3 – Erreur absolue L^2 par bande de s (norme discrète pondérée par l'aire des cellules).

3 Grecques au strike

Source : data/evaluate_greeks_no_corner/20260921_corner_treatments_iters50000. Référence : dérivée symbolique (sympy) de la forme fermée de Reiner–Rubinstein.

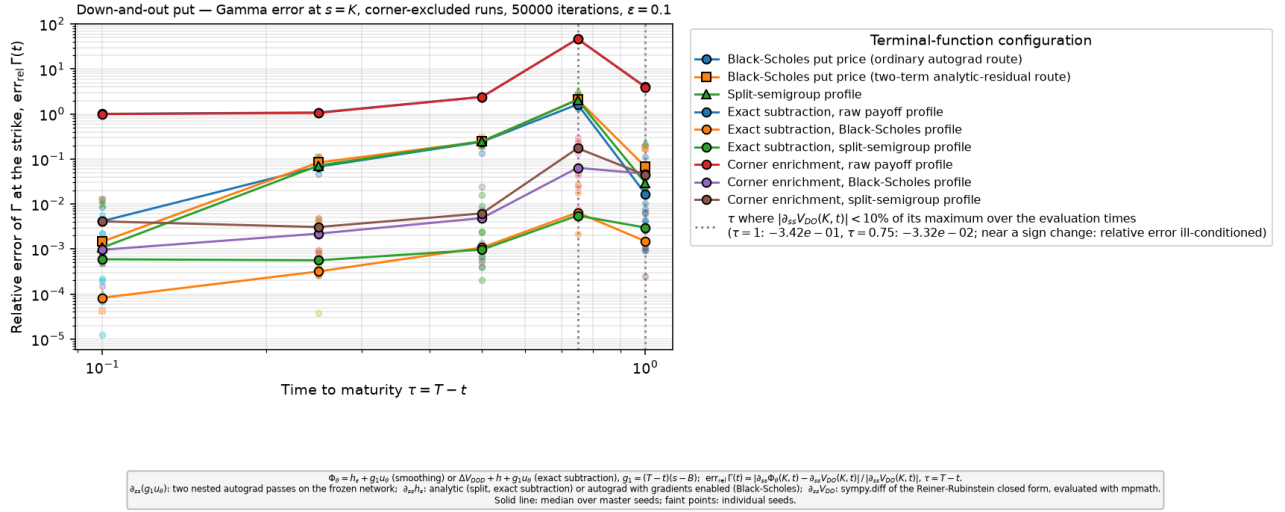


FIGURE 4 – Erreur relative sur $\Gamma(K, t) = \partial_{ss} \Phi_\theta(K, t)$ en fonction de $\tau = T - t$, une courbe par configuration (médiane sur les graines ; points : graines).

Configuration	t	tau = T - t	Delta exact	Gamma exact	n seeds	err_rel_Delta	err_rel_Gamma
Black-Scholes put price (ordinary autograd route)	0	1	-1.4818e-01	-3.4150e-01 (*)	5	1.064e-01 [6.965e-02, 1.167e-01]	1.679e-02 [4.303e-03, 1.986e-01]
Black-Scholes put price (ordinary autograd route)	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	1.036e-01 [8.758e-02, 1.593e-01]	1.640e+00 [1.315e+00, 2.027e+00]
Black-Scholes put price (ordinary autograd route)	0.5	0.5	-3.4265e-01	+7.9222e-01	5	8.379e-02 [7.063e-02, 1.008e-01]	2.416e-01 [1.368e-01, 2.757e-01]
Black-Scholes put price (ordinary autograd route)	0.75	0.25	-4.4335e-01	+2.4764e+00	5	2.133e-02 [1.585e-02, 4.228e-02]	6.750e-02 [4.648e-02, 1.109e-01]
Black-Scholes put price (ordinary autograd route)	0.9	0.1	-4.6849e-01	+4.1920e+00	5	2.612e-03 [9.225e-04, 1.108e-02]	4.180e-03 [3.842e-03, 1.333e-02]
Black-Scholes put price (two-term analytic-residual route)	0	1	-1.4818e-01	-3.4150e-01 (*)	5	1.355e-01 [7.518e-02, 1.415e-01]	6.690e-02 [2.020e-02, 1.776e-01]
Black-Scholes put price (two-term analytic-residual route)	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	1.071e-01 [8.129e-02, 1.165e-01]	2.104e+00 [1.778e+00, 2.610e+00]
Black-Scholes put price (two-term analytic-residual route)	0.5	0.5	-3.4265e-01	+7.9222e-01	5	8.190e-02 [6.495e-02, 8.626e-02]	2.449e-01 [2.016e-01, 3.001e-01]
Black-Scholes put price (two-term analytic-residual route)	0.75	0.25	-4.4335e-01	+2.4764e+00	5	2.621e-02 [2.055e-02, 3.734e-02]	8.384e-02 [6.869e-02, 1.122e-01]
Black-Scholes put price (two-term analytic-residual route)	0.9	0.1	-4.6849e-01	+4.1920e+00	5	1.601e-03 [4.936e-05, 6.331e-03]	1.467e-03 [4.283e-05, 1.203e-02]
Split-semigroup profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	1.003e-01 [1.493e-02, 1.438e-01]	2.924e-02 [2.166e-02, 2.517e-01]
Split-semigroup profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	9.237e-02 [4.905e-02, 1.468e-01]	2.068e+00 [1.823e+00, 3.518e+00]
Split-semigroup profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	7.916e-02 [5.065e-02, 1.251e-01]	2.490e-01 [1.980e-01, 2.781e-01]
Split-semigroup profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	2.505e-02 [1.876e-02, 2.705e-02]	7.167e-02 [6.584e-02, 1.075e-01]
Split-semigroup profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	4.039e-04 [2.811e-04, 2.158e-03]	1.078e-03 [1.043e-04, 1.212e-02]
Exact subtraction, raw payoff profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	3.228e+00 [3.225e+00, 3.240e+00]	4.017e+00 [3.574e+00, 4.073e+00]
Exact subtraction, raw payoff profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	2.088e+00 [2.085e+00, 2.095e+00]	4.681e+01 [4.250e+01, 4.806e+01]
Exact subtraction, raw payoff profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	1.416e+00 [1.413e+00, 1.418e+00]	2.411e+00 [2.150e+00, 2.451e+00]
Exact subtraction, raw payoff profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	1.103e+00 [1.101e+00, 1.105e+00]	1.083e+00 [9.903e-01, 1.099e+00]
Exact subtraction, raw payoff profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	1.052e+00 [1.050e+00, 1.053e+00]	1.008e+00 [9.435e-01, 1.019e+00]
Exact subtraction, Black-Scholes profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	1.549e-04 [8.349e-05, 6.463e-04]	1.490e-03 [2.459e-04, 6.698e-03]
Exact subtraction, Black-Scholes profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	1.472e-04 [1.853e-06, 6.461e-04]	6.462e-03 [6.078e-03, 2.409e-02]
Exact subtraction, Black-Scholes profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	2.426e-04 [5.164e-06, 3.615e-04]	1.076e-03 [5.006e-04, 1.351e-03]
Exact subtraction, Black-Scholes profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	7.842e-05 [2.869e-05, 3.811e-04]	3.171e-04 [3.742e-05, 4.418e-04]
Exact subtraction, Black-Scholes profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	7.921e-05 [2.017e-05, 1.380e-04]	8.269e-05 [1.223e-05, 2.208e-04]
Exact subtraction, split-semigroup profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	3.002e-04 [1.269e-04, 7.926e-04]	2.990e-03 [9.200e-04, 1.035e-02]
Exact subtraction, split-semigroup profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	1.656e-04 [3.019e-05, 4.315e-04]	5.542e-03 [2.196e-03, 5.806e-02]
Exact subtraction, split-semigroup profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	2.074e-04 [1.060e-04, 5.968e-04]	9.687e-04 [3.884e-04, 2.451e-03]
Exact subtraction, split-semigroup profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	1.640e-04 [4.401e-05, 3.802e-04]	5.615e-04 [3.282e-04, 9.357e-04]
Exact subtraction, split-semigroup profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	1.923e-04 [9.877e-05, 5.444e-04]	5.914e-04 [1.495e-04, 1.274e-03]
Corner enrichment, raw payoff profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	3.232e+00 [3.215e+00, 3.246e+00]	3.929e+00 [3.583e+00, 4.319e+00]
Corner enrichment, raw payoff profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	2.092e+00 [2.080e+00, 2.101e+00]	4.672e+01 [4.584e+01, 4.743e+01]
Corner enrichment, raw payoff profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	1.418e+00 [1.411e+00, 1.420e+00]	2.396e+00 [2.259e+00, 2.438e+00]
Corner enrichment, raw payoff profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	1.105e+00 [1.104e+00, 1.106e+00]	1.060e+00 [1.032e+00, 1.081e+00]
Corner enrichment, raw payoff profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	1.052e+00 [1.041e+00, 1.054e+00]	9.978e-01 [9.804e-01, 1.018e+00]
Corner enrichment, Black-Scholes profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	1.388e-03 [6.775e-04, 1.661e-03]	4.714e-02 [7.676e-03, 1.132e-01]
Corner enrichment, Black-Scholes profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	7.944e-04 [1.927e-05, 1.635e-03]	6.375e-02 [1.802e-02, 2.470e-01]
Corner enrichment, Black-Scholes profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	1.193e-03 [5.798e-05, 1.880e-03]	4.873e-03 [2.039e-04, 1.618e-02]
Corner enrichment, Black-Scholes profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	7.177e-04 [2.937e-04, 1.987e-03]	2.198e-03 [2.639e-04, 4.902e-03]
Corner enrichment, Black-Scholes profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	6.028e-04 [3.209e-05, 7.496e-04]	9.626e-04 [7.398e-04, 1.694e-03]
Corner enrichment, split-semigroup profile	0	1	-1.4818e-01	-3.4150e-01 (*)	5	3.539e-03 [1.628e-03, 6.720e-03]	4.405e-02 [1.123e-03, 7.510e-02]
Corner enrichment, split-semigroup profile	0.25	0.75	-2.3063e-01	-3.3151e-02 (*)	5	8.760e-04 [9.496e-05, 3.735e-03]	1.740e-01 [4.253e-02, 3.060e-01]
Corner enrichment, split-semigroup profile	0.5	0.5	-3.4265e-01	+7.9222e-01	5	8.006e-04 [2.709e-04, 2.185e-03]	6.199e-03 [4.043e-04, 2.404e-02]
Corner enrichment, split-semigroup profile	0.75	0.25	-4.4335e-01	+2.4764e+00	5	6.953e-04 [2.344e-04, 2.595e-03]	3.079e-03 [6.961e-04, 4.406e-03]
Corner enrichment, split-semigroup profile	0.9	0.1	-4.6849e-01	+4.1920e+00	5	1.589e-03 [1.576e-04, 1.871e-03]	4.114e-03 [2.167e-03, 5.759e-03]

TABLE 4 – Δ et Γ au strike : erreur relative ponctuelle, médiane [min, max] sur les graines ; (*) marque un Γ exact proche d'un changement de signe.

4 Figures par run (graine 0)

4.1 Black-Scholes put price (ordinary autograd route)

lissage du coin, profil Black-Scholes V^e , route autograd ordinaire. Run : 20260913_045605_iters50000_eps0.1_seed0_blacksholes_nocorner.

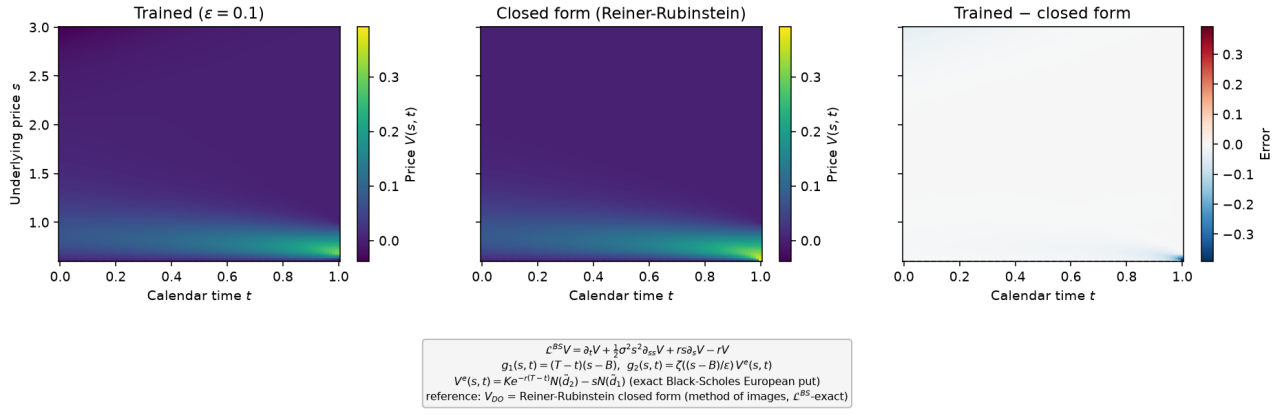


FIGURE 5 – Surface de prix : solution entraînée, forme fermée, différence. lissage du coin, profil Black-Scholes V^e , route autograd ordinaire.

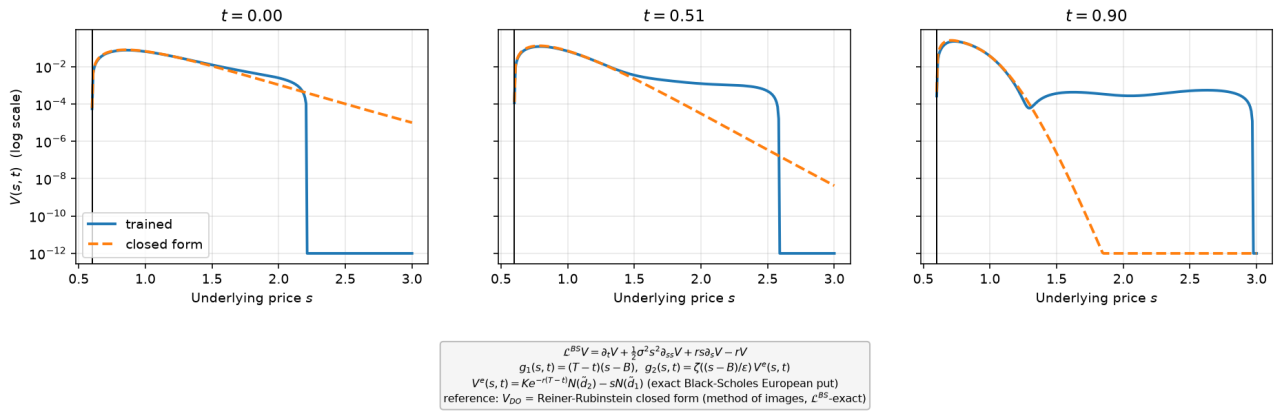


FIGURE 6 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). lissage du coin, profil Black-Scholes V^e , route autograd ordinaire.

4.2 Black-Scholes put price (two-term analytic-residual route)

lissage du coin, profil Black-Scholes V^e , route analytique à deux termes. Run : 20260913_035016_iters50000_eps0.1_seed0_blacksholes_analyticres_nocorner.

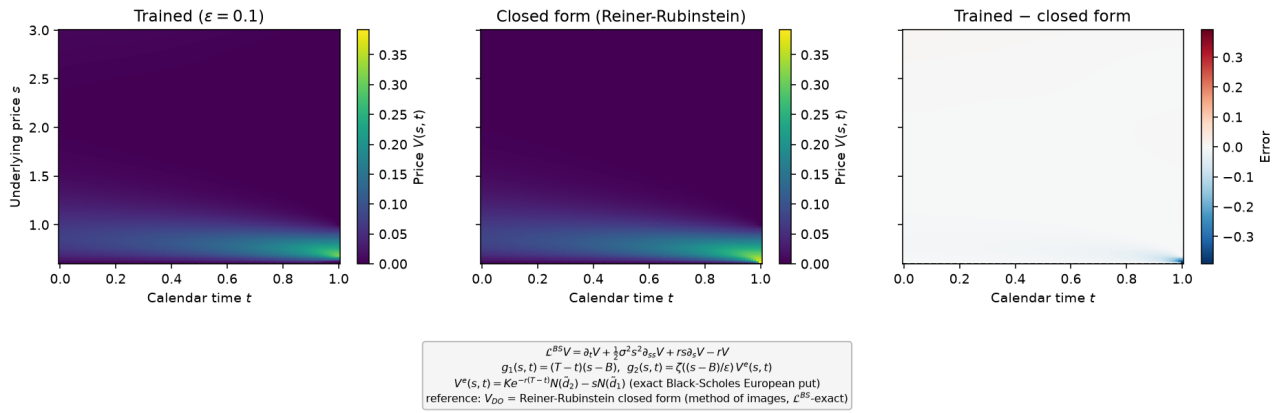


FIGURE 7 – Surface de prix : solution entraînée, forme fermée, différence. lissage du coin, profil Black-Scholes V^e , route analytique à deux termes.

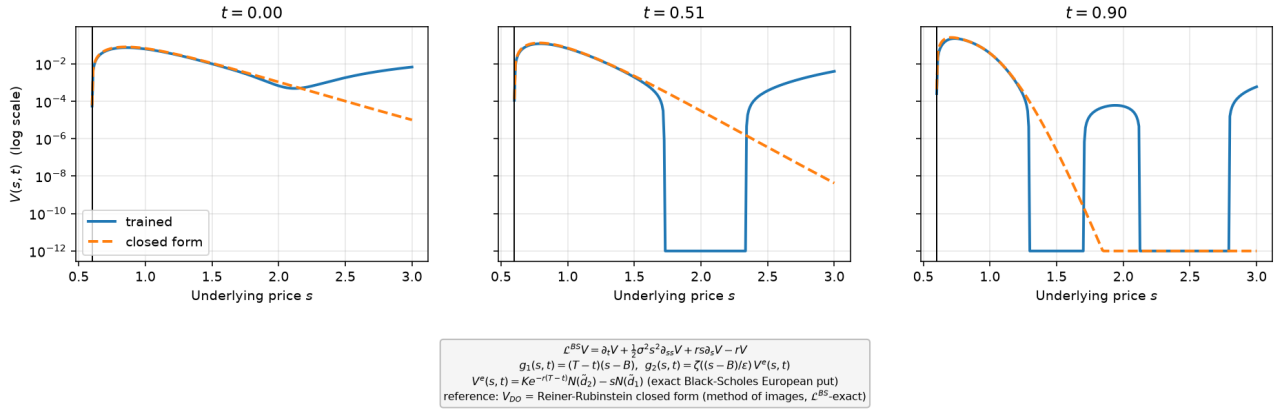


FIGURE 8 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). lissage du coin, profil Black-Scholes V^e , route analytique à deux termes.

4.3 Split-semigroup profile

lissage du coin, profil split-semigroupe. Run : 20260912_071515_iters50000_eps0.1_seed0_split_nuc0.3_nquad8000_nocorner.

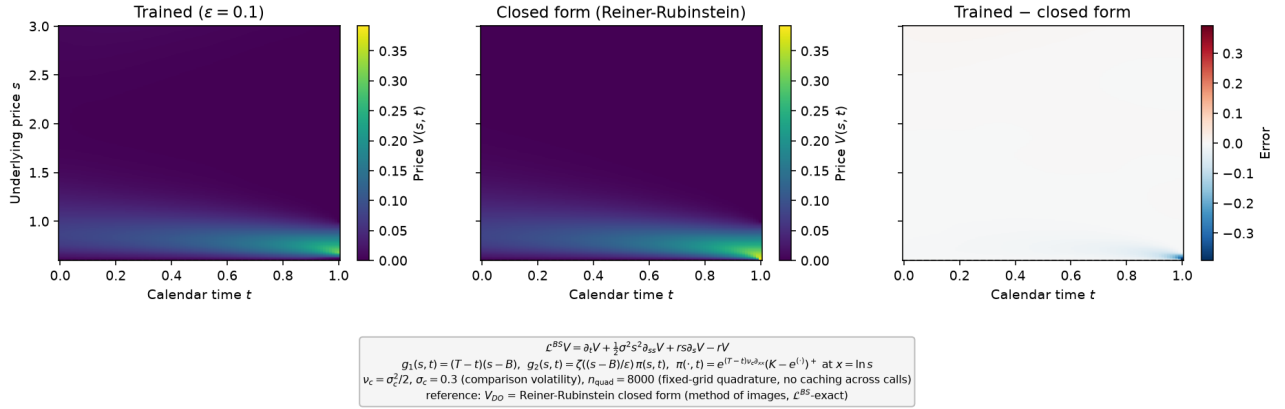


FIGURE 9 – Surface de prix : solution entraînée, forme fermée, différence. lissage du coin, profil split-semigroupe.

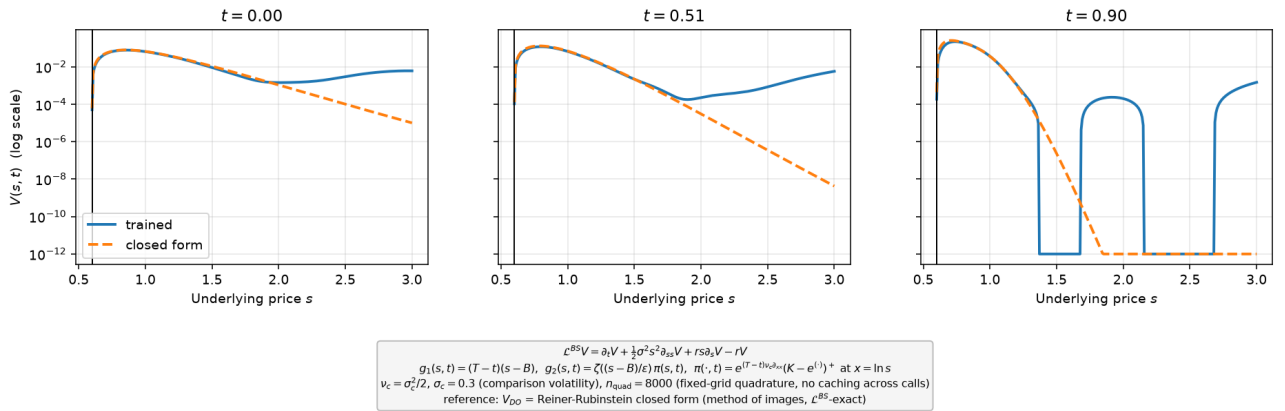


FIGURE 10 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). lissage du coin, profil split-semigroupe.

4.4 Exact subtraction, raw payoff profile

soustraction exacte (ΔV_{DOD}), profil payoff brut (contrôle négatif). Run : 20260920_201745_itsers50000_eps0_seed0_subtraction_raw.

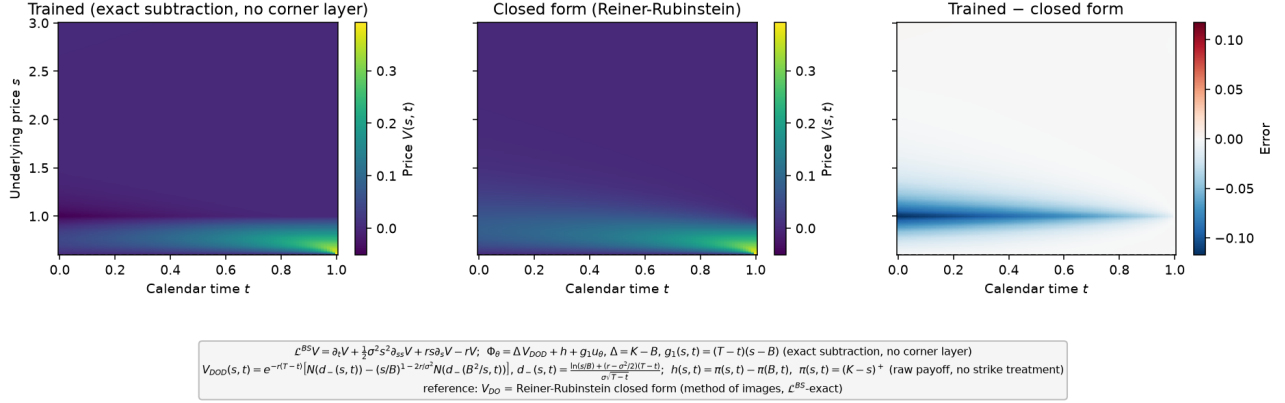


FIGURE 11 – Surface de prix : solution entraînée, forme fermée, différence. soustraction exacte (ΔV_{DOD}), profil payoff brut (contrôle négatif).

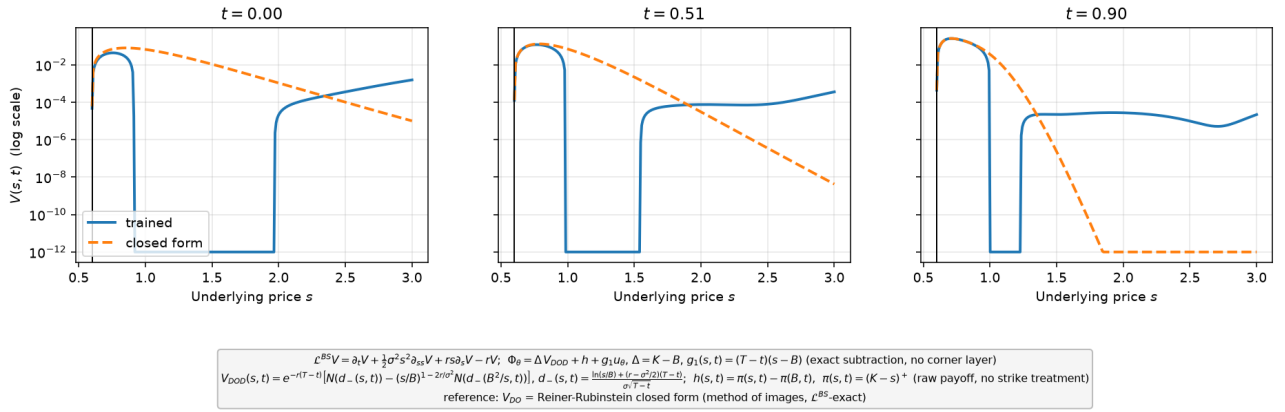


FIGURE 12 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). soustraction exacte (ΔV_{DOD}), profil payoff brut (contrôle négatif).

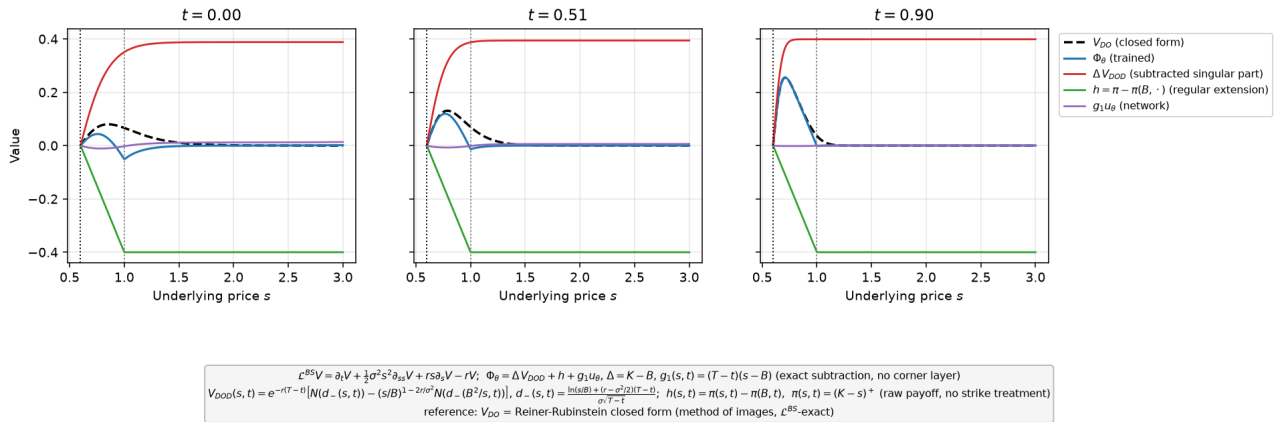


FIGURE 13 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . soustraction exacte (ΔV_{DOD}), profil payoff brut (contrôle négatif).

4.5 Exact subtraction, Black-Scholes profile

soustraction exacte (ΔV_{DOD}), profil Black-Scholes V^e . Run : 20260920_201746_iters50000_eps0_seed0_subtraction_blackscholes.

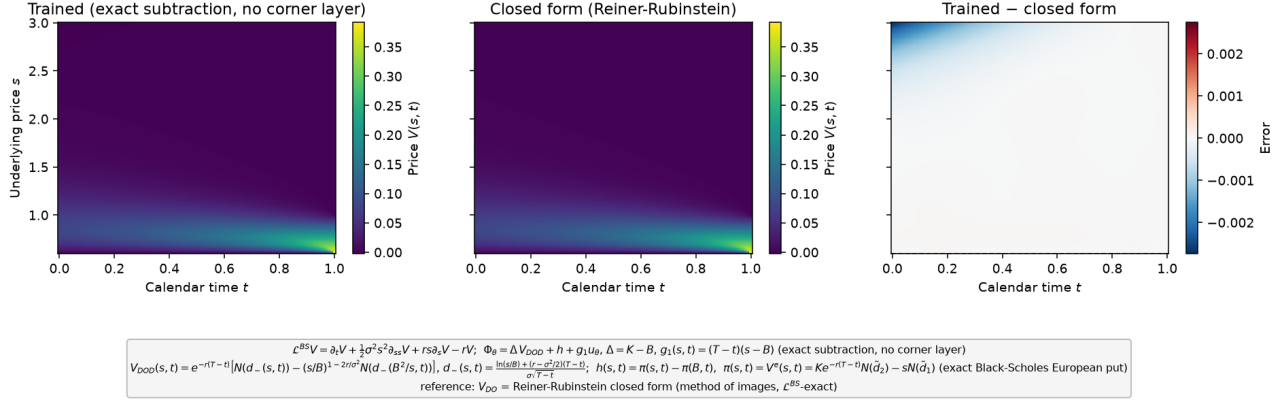


FIGURE 14 – Surface de prix : solution entraînée, forme fermée, différence. soustraction exacte (ΔV_{DOD}), profil Black-Scholes V^e .

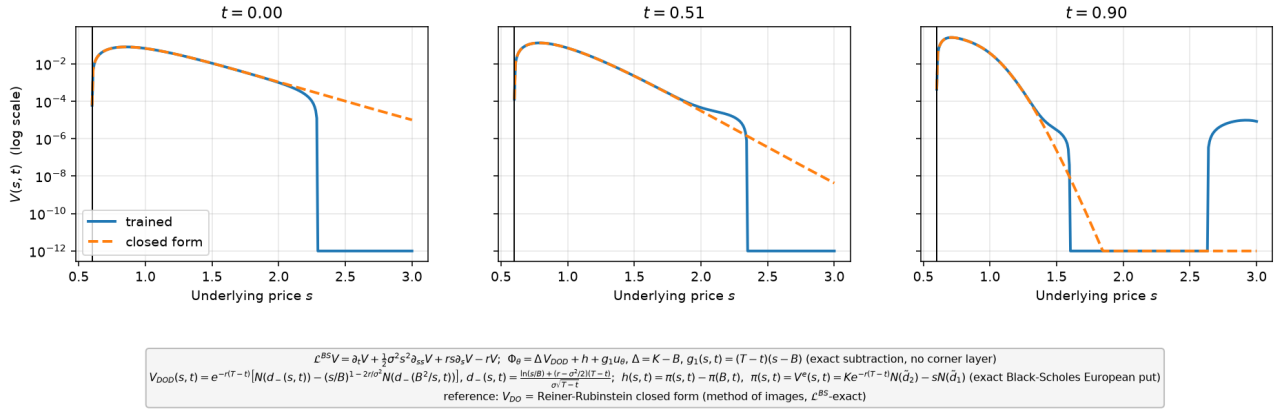


FIGURE 15 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). soustraction exacte (ΔV_{DOD}), profil Black-Scholes V^e .

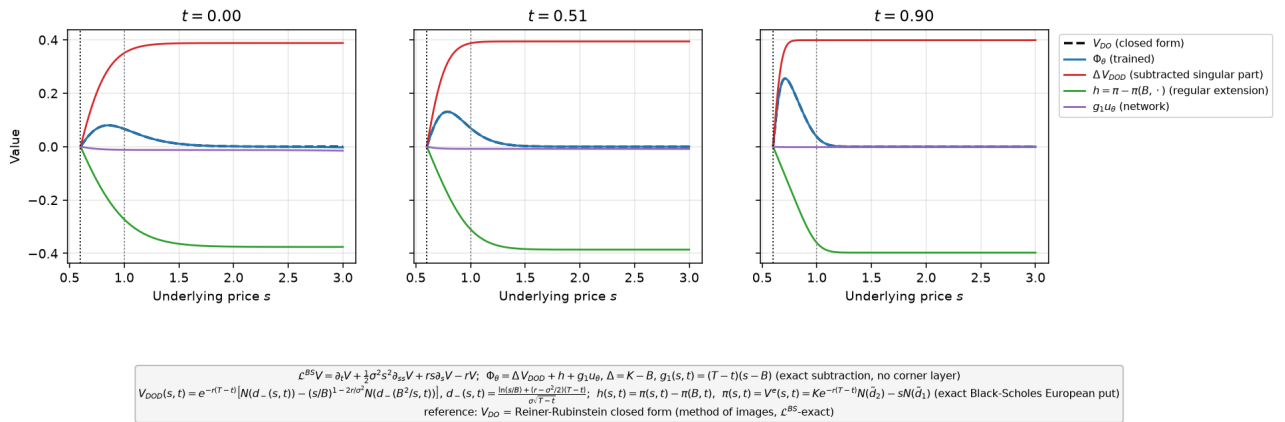


FIGURE 16 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . soustraction exacte (ΔV_{DOD}), profil Black-Scholes V^e .

4.6 Exact subtraction, split-semigroup profile

soustraction exacte (ΔV_{DOD}), profil split-semigroupe. Run : 20260920_210040_iters50000_eps0_seed0_subtraction_split_nuc0.3_closedform.

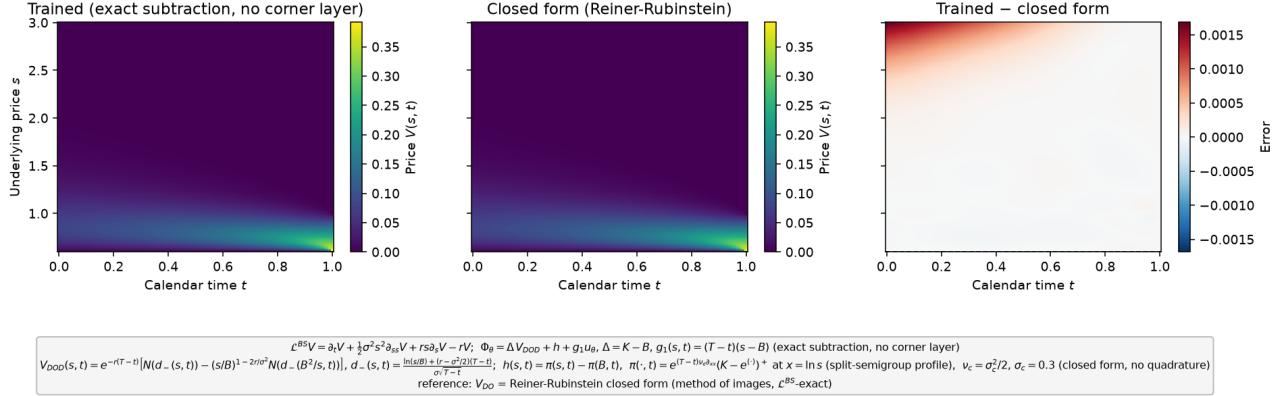


FIGURE 17 – Surface de prix : solution entraînée, forme fermée, différence. soustraction exacte (ΔV_{DOD}), profil split-semigroupe.

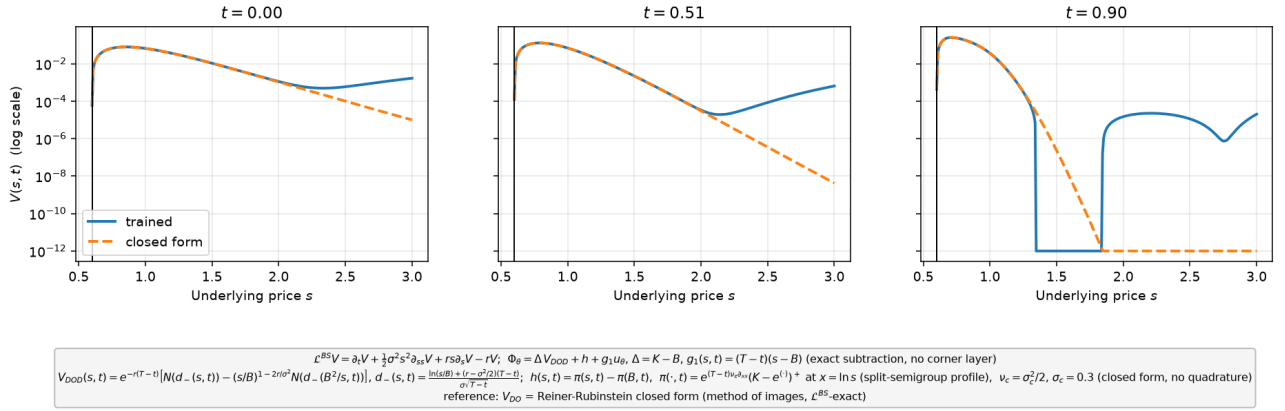


FIGURE 18 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). soustraction exacte (ΔV_{DOD}), profil split-semigroupe.

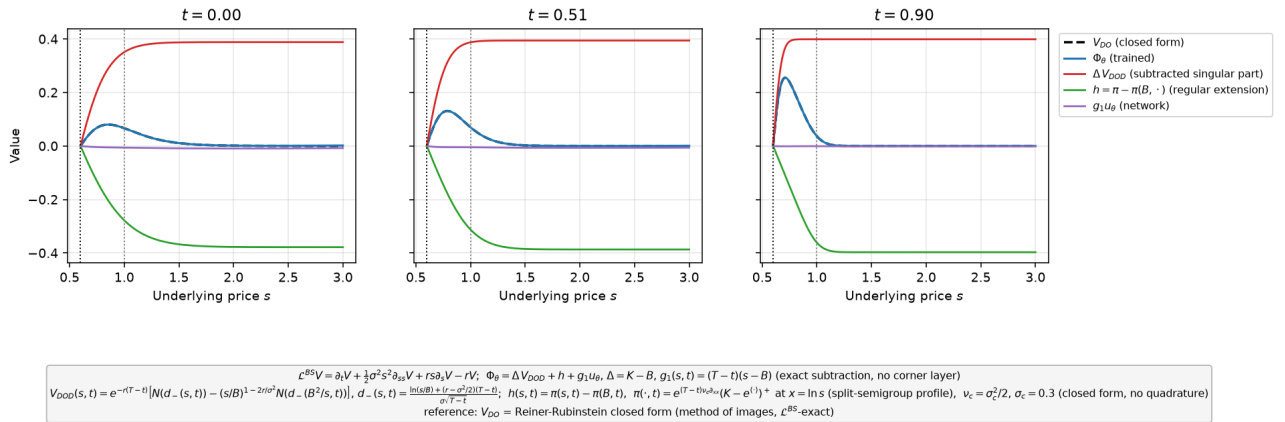


FIGURE 19 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . soustraction exacte (ΔV_{DOD}), profil split-semigroupe.

4.7 Corner enrichment, raw payoff profile

enrichissement du coin ($\chi \Delta \text{erf}(\xi)$), profil payoff brut (contrôle négatif). Run : 20260921_003020_its50000_eps0_seed0_enrichment_raw_d00.1_d10.3.

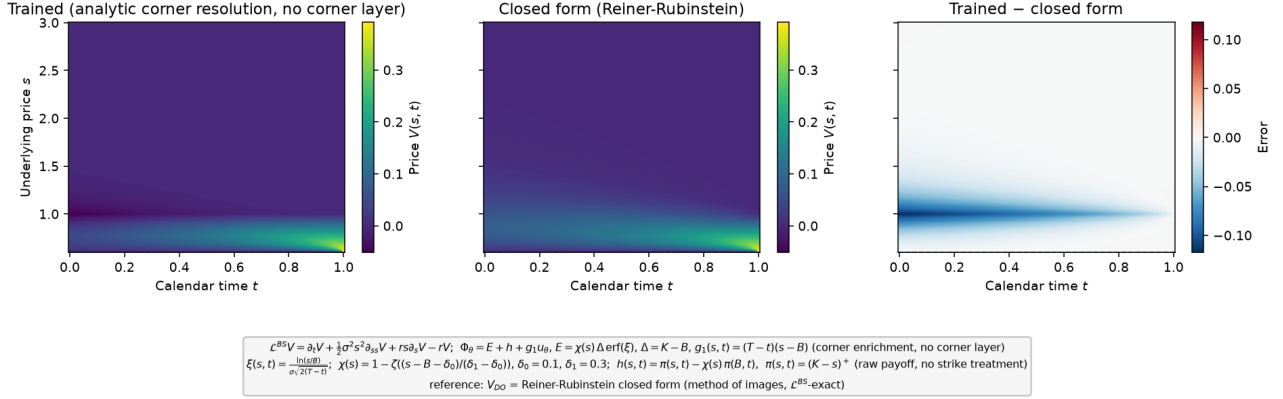


FIGURE 20 – Surface de prix : solution entraînée, forme fermée, différence. enrichissement du coin ($\chi \Delta \text{erf}(\xi)$), profil payoff brut (contrôle négatif).

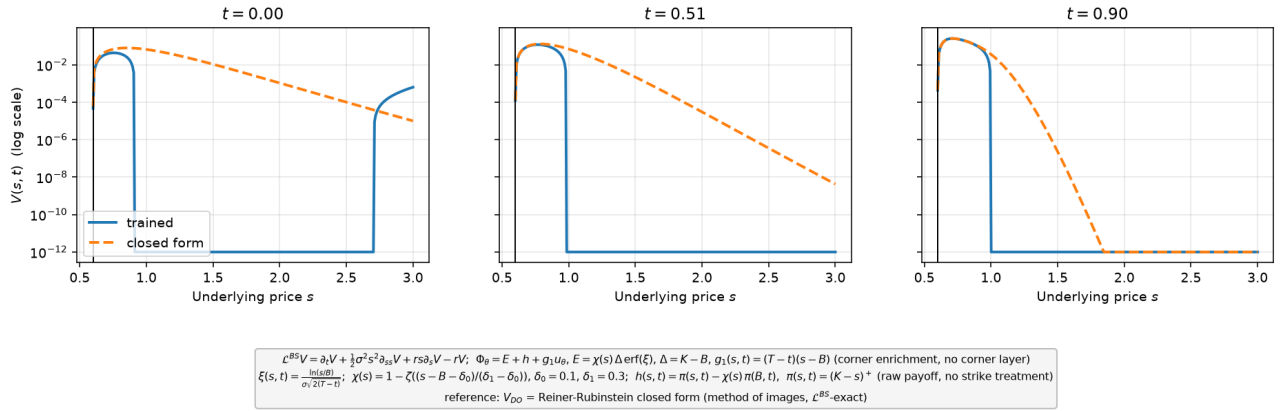


FIGURE 21 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). enrichissement du coin ($\chi \Delta \text{erf}(\xi)$), profil payoff brut (contrôle négatif).

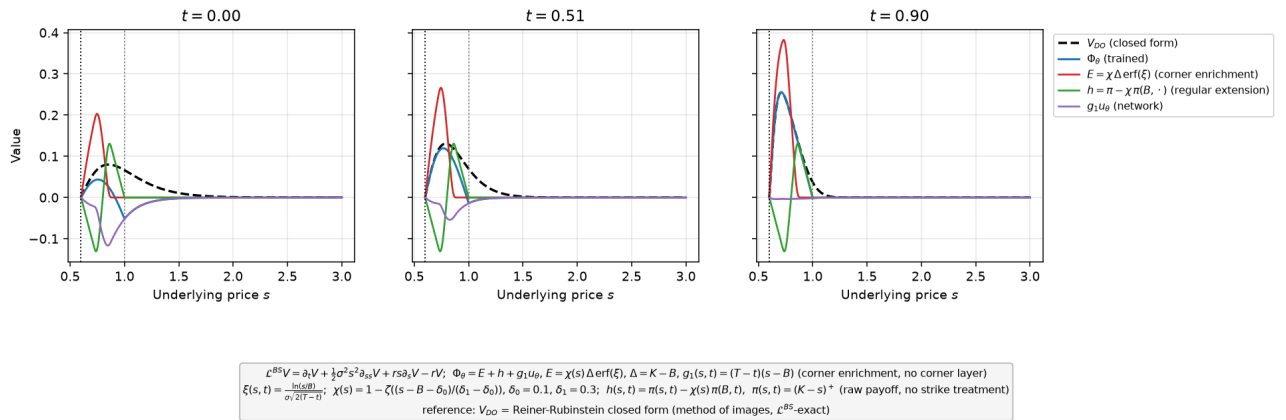


FIGURE 22 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . enrichissement du coin ($\chi \Delta \text{erf}(\xi)$), profil payoff brut (contrôle négatif).

4.8 Corner enrichment, Black-Scholes profile

enrichissement du coin ($\chi\Delta \operatorname{erf}(\xi)$), profil Black-Scholes V^e . Run : 20260921_003313_iters50000_eps0_seed0_enrichment_blacksholes_d00.1_d10.3.

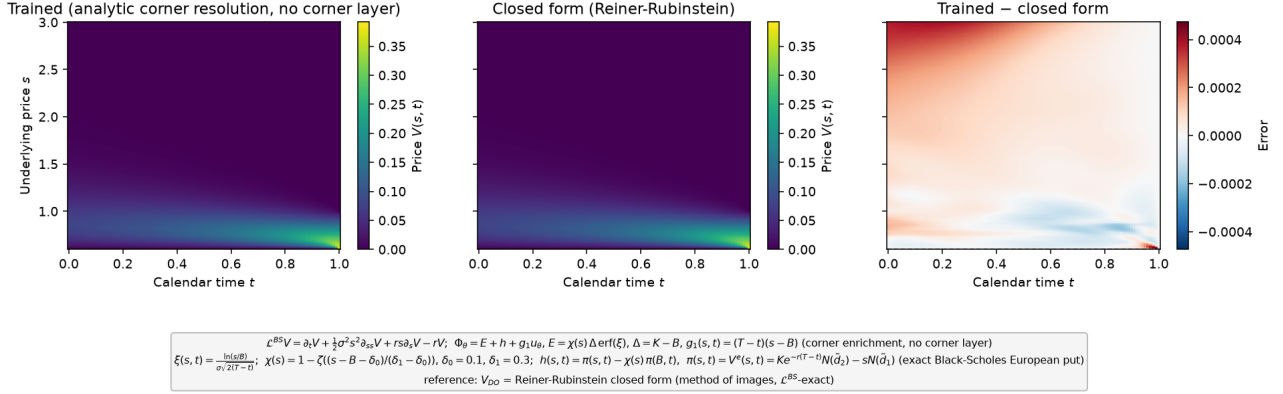


FIGURE 23 – Surface de prix : solution entraînée, forme fermée, différence. enrichissement du coin ($\chi\Delta \operatorname{erf}(\xi)$), profil Black-Scholes V^e .

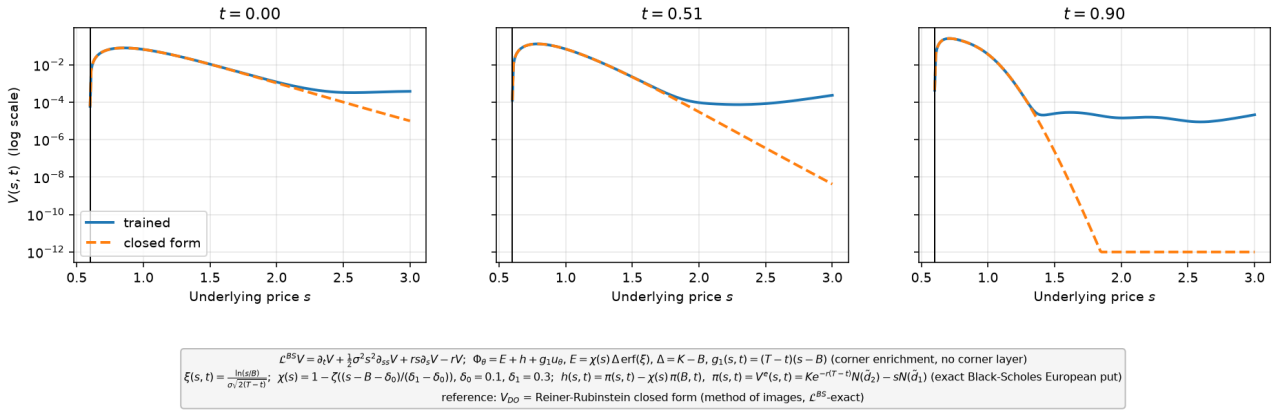


FIGURE 24 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). enrichissement du coin ($\chi\Delta \operatorname{erf}(\xi)$), profil Black-Scholes V^e .

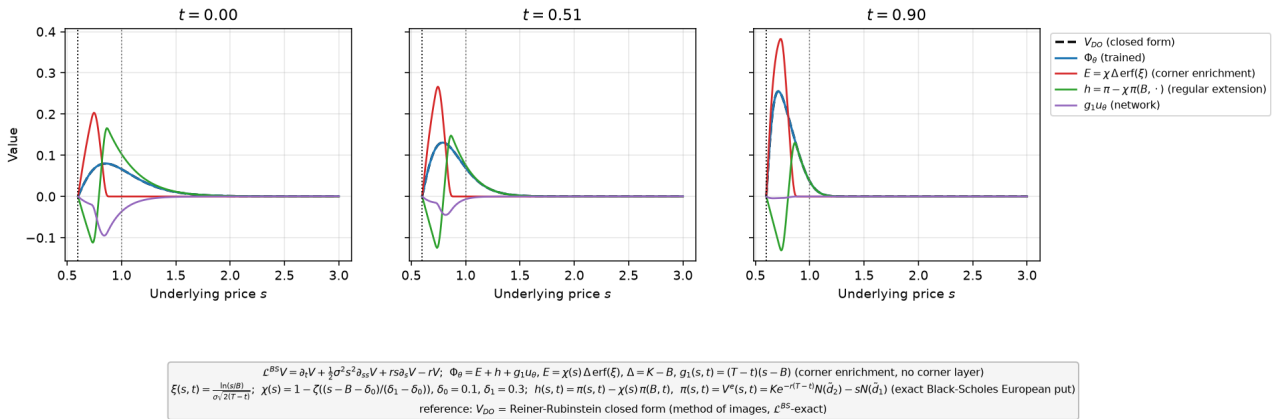


FIGURE 25 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . enrichissement du coin ($\chi\Delta \operatorname{erf}(\xi)$), profil Black-Scholes V^e .

4.9 Corner enrichment, split-semigroup profile

enrichissement du coin ($\chi\Delta\text{erf}(\xi)$), profil split-semigroupe. Run : 20260921_011735_iters50000_eps0_seed0_enrichment_split_d00.1_d10.3_nuc0.3_closedform.

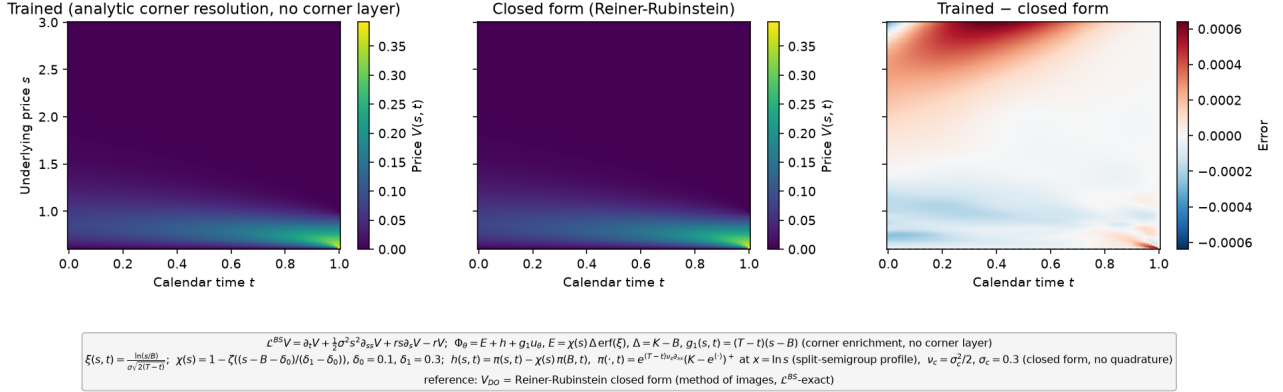


FIGURE 26 – Surface de prix : solution entraînée, forme fermée, différence. enrichissement du coin ($\chi\Delta\text{erf}(\xi)$), profil split-semigroupe.

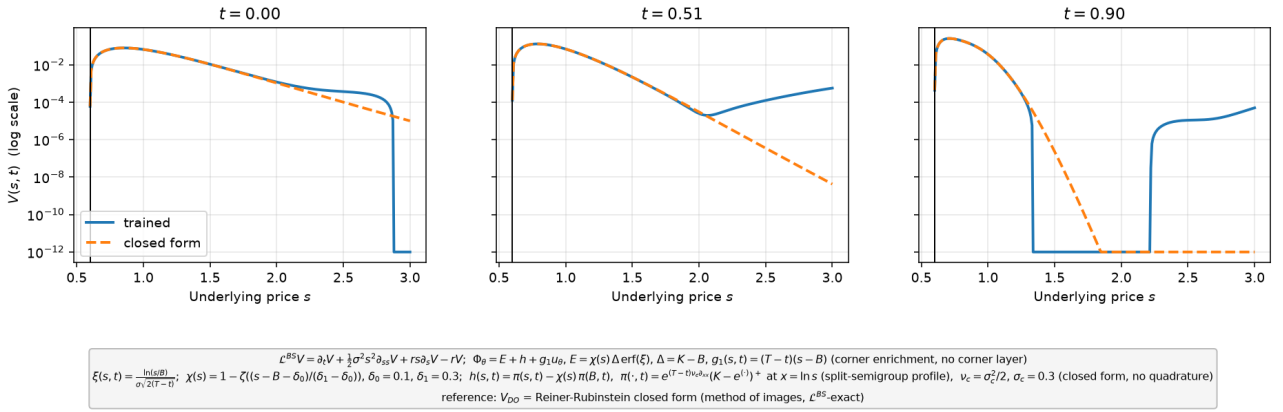


FIGURE 27 – Coupes $V(s, t)$ à t fixé, échelle logarithmique (solution entraînée en trait plein, forme fermée en tirets). enrichissement du coin ($\chi\Delta\text{erf}(\xi)$), profil split-semigroupe.

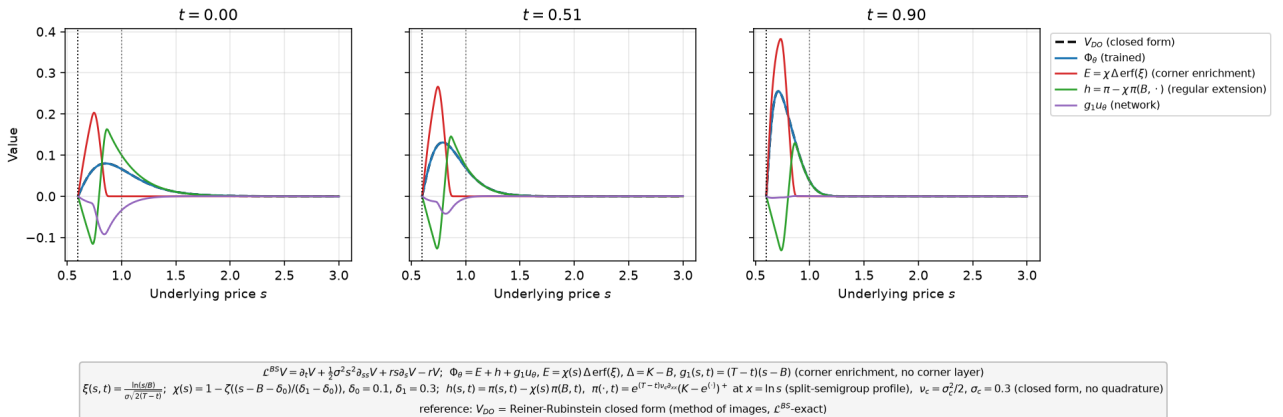


FIGURE 28 – Décomposition de l'estimateur : partie singulière en forme fermée, extension régulière h , réseau $g_1 u_\theta$, total, contre V_{DO} . enrichissement du coin ($\chi\Delta\text{erf}(\xi)$), profil split-semigroupe.